5.3 Solving Systems of Linear Equations by Elimination

Name _____ Period ____ Date _____

Algebra I 5.3 · Student sheet

I can…

- solve systems by adding or subtracting equations.
- solve systems by multiplying first, then eliminating.
- solve real-life problems using elimination.

Vocabulary · fill in the missing word

Elimination — add or subtract the _____ so that one variable cancels

Opposite coefficients — _____

Part 1 · Make it cancel

System	Add, subtract, or multiply first? (by what?)	Solution
$x + y = 6$; $x - y = 2$		
$3x + 2y = 7$; $3x - y = 1$		
$x + 2y = 5$; $3x - y = 1$		
$2x + 3y = 8$; $5x - 2y = 1$		

Part 2 · Notes and examples (fill in each note, then work the example)

Add or subtract?: Add when the coefficients are _____. Subtract when they are _____.

Example 1

Solve.

$3x + 2y = 4$

$5x - 2y = 12$

Multiply first: If no coefficients match, _____ one or both equations first.

Careful: Multiply _____ term in the equation.

Example 2

Solve.

$2x + 3y = 1$

$x + 2y = 0$

Example 3

Solve.

$4x + 3y = 10$

$3x - 2y = -1$

Example 4

2 pizzas and 3 salads cost $38. 4 pizzas and 1 salad cost $56. Find the cost of each.

Part 3 · Partner practice

Partner 1

Solve: $2x + y = 9$ and $2x - 3y = 1$.

Partner 2

Solve: $x + 3y = 7$ and $2x - y = 0$.